# Questions: Introduction to rearranging equations

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**Summary**

A selection of questions for the study guide on introduction to rearranging equations.

*Before attempting these questions, it is highly recommended that you read Guide: Introduction to rearranging equations.*

## Q1

For each of the following equations, rearrange the equation for the variable given.

1.1. Rearrange $x = a + 2b$ for $a$.

1.2. Rearrange $x = a + 2b$ for $b$.

1.3. Rearrange $x - 2y + 4z = 4$ for $z$.

1.4. Rearrange $5x - 3y + 8z = -2$ for $x$.

1.5. Rearrange $5x - 3y + 8z = -2$ for $y$.

1.6. Rearrange $5x - 3y + 8z = -2$ for $z$.

1.7. Rearrange $x^2 + y^2 = 4$ for $x$.

1.8. Rearrange $\frac{x^2}{4} + \frac{y^2}{16} = a$ for $x$.

1.9. Rearrange $\frac{x^2}{4} + \frac{y^2}{16} = a$ for $y$.

1.10. Rearrange $\sqrt{x^2 - a^2} = y + 1$ for $x$.

1.11. Rearrange $\sqrt[3]{x^3 - a^3} = y + 1$ for $a$.

1.12. Rearrange $\sqrt[3]{x^3 - a^3} = y + 1$ for $x$.

1.13. Rearrange $x^4y^2 = a^3 + 2bcd$ for $d$.

1.14. Rearrange $x^4y^2 = a^3 + 2bcd$ for $a$.

1.15. Rearrange $x^4y^2 = a^3 + 2bcd$ for $x$.

1.16. Rearrange $\frac{1}{x} + 45 = ly^2$ for $x$.

## Q2

In Guide: Introduction to rearranging equations, you saw the expression

$$5x^3y^3 + \frac{6z}{w^4} = 4abc^2$$

where you rearranged this equation for $x$.

Rearrange this expression for every other variable $a, b, c, y, z, w$.

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After attempting the questions above, please click this link to find the answers.

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## Version history and licensing

v1.0: initial version created 08/23 by Shanelle Advani and tdhc as part of a University of St Andrews STEP project.

- v1.1: edited 05/24 by tdhc.

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